

Polya Noise-Erase Channel (NEC)

$$Y_k = \overset{\text{erasure}}{\underset{\substack{\uparrow \\ \text{output RV} \\ \in \{0,1,E\}}}{E}} \cdot \overset{\text{indicator function}}{\underset{\substack{\uparrow \\ \text{noise-erasure} \\ \text{RV} \in \{0,1,E\}}}{1\{Z_k=E\}}} + (\overset{\text{input}}{\underset{\substack{\uparrow \\ \in \{0,1\}}}{X_k} \oplus Z_k) \cdot 1\{Z_k \neq E\}}, \quad k=1,2,\dots$$

$$\text{w/ } Z_k \perp\!\!\!\perp X_k \quad \forall k, \ell$$

$$\Rightarrow Z_k = \underset{\substack{\uparrow \\ \text{output RV} \\ \in \{0,1,E\}}}{E} \cdot 1\{Y_k=E\} + (Y_k \oplus \underset{\substack{\uparrow \\ \text{input} \\ \in \{0,1\}}}{X_k}) \cdot 1\{Y_k \neq E\} \quad \oplus$$

Want to derive $P(Y^m = b(j) | X^m = b(i))$

where $Y^m = (Y_0, Y_1, \dots, Y_{m-1})$, $X^m = (X_0, X_1, \dots, X_{m-1})$,

$b(j) = (b_{j,0}, b_{j,1}, \dots, b_{j,m-1})$, $b(i) = (b_{i,0}, b_{i,1}, \dots, b_{i,m-1})$.

Know: $\{Z_i\}$ is a stationary Markov chain (of Memory $M=1$) with

$$\begin{aligned}
 P(Z_k = z) &= \overset{1\{z=E\}}{\varepsilon'} \cdot \overset{1\{z=1\}}{\varepsilon} \cdot \overset{1\{z=0\}}{(1-\varepsilon-\varepsilon')} \\
 &= \begin{cases} \varepsilon' & \text{if } z=E \\ \varepsilon & \text{if } z=1 \\ 1-\varepsilon-\varepsilon' & \text{if } z=0 \end{cases}
 \end{aligned}$$

where $0 \leq \varepsilon + \varepsilon' < 1$.

$P(Z_k = a_k | Z_{k-1} = a_{k-1})$ is given by this transition matrix

$$\mathcal{Q} = [P_{a_{k-1}, a_k}] = \begin{matrix} \begin{matrix} 0 & 1 & E \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ E \end{matrix} & \begin{bmatrix} \frac{1-\varepsilon-\varepsilon'+\delta}{1+\delta} & \frac{\varepsilon}{1+\delta} & \frac{\varepsilon'}{1+\delta} \\ \frac{1-\varepsilon-\varepsilon'}{1+\delta} & \frac{\varepsilon+\delta}{1+\delta} & \frac{\varepsilon'}{1+\delta} \\ \frac{1-\varepsilon-\varepsilon'}{1+\delta} & \frac{\varepsilon}{1+\delta} & \frac{\varepsilon+\delta}{1+\delta} \end{bmatrix} \end{matrix}$$

s_0

$$P(Z_k = a_k | Z_{k-1} = a_{k-1})$$

$$= \frac{1}{1+\delta} \left[\varepsilon' + 1\{a_{k-1} = E\} \delta \right] \cdot \left[\varepsilon + 1\{a_{k-1} = I\} \delta \right] \cdot \left[1 - \varepsilon - \varepsilon' + 1\{a_{k-1} = 0\} \delta \right]$$

Hence

$$P(Y^m = b(j) | X^m = b(i)) = P(Y_0 = b_{j,0}, \dots, Y_{m-1} = b_{j,m-1} | X_0 = b_{i,0}, \dots, X_{m-1} = b_{i,m-1})$$

by ④

$$\downarrow = P(Z_0 = s_0, Z_1 = s_1, \dots, Z_m = s_m) \text{ where } s_k = E \cdot 1\{b_{j,k} = E\} + (b_{j,k} \oplus b_{i,k}) \cdot 1\{b_{j,k} \neq E\}$$

$$= P(Z_0 = s_0) \prod_{k=1}^{m-1} P(Z_k = s_k | Z_{k-1} = s_{k-1}) \leftarrow \{Z_k\} \text{ Markov}$$

$$= \left[\varepsilon' \cdot 1\{b_{j,0} = E\} + \begin{pmatrix} b_{j,0} \oplus b_{i,0} & 1 - b_{j,0} \oplus b_{i,0} \\ \varepsilon & 1 - \varepsilon' - \varepsilon \end{pmatrix} 1\{b_{j,0} \neq E\} \right] \\ \prod_{k=1}^{m-1} \left[\frac{\varepsilon' + 1\{b_{j,k-1} = E\} \delta}{1 + \delta} \right] \left[\frac{\varepsilon + 1\{b_{j,k-1} \oplus b_{i,k-1} = I\} \delta}{1 + \delta} \right] \left[\frac{1 - \varepsilon' - \varepsilon + 1\{b_{j,k-1} \oplus b_{i,k-1} = 0\} \delta}{1 + \delta} \right]$$